

§2.4. compactness and properness

prop: $X(\Delta)$ is compact if and only if $|\Delta| = N_R$

Notion: a complex variety is ~~compact~~ in classical topology when it is complete (proper) as an algebraic ~~geometric~~ variety.

Notion: proper map.

$f: X \rightarrow Y$ is proper if the preimage of compact set in Y is compact in X

Remark: X is compact $\Leftrightarrow$ the constant map $X \rightarrow Y = \{\text{a single point}\}$
 $\varphi: N' \rightarrow N$ is compatible is proper

prop: The map $\varphi_*: X(\Delta') \rightarrow X(\Delta)$ is proper if and only if $\varphi^{-1}(|\Delta|) = |\Delta'|$

prop: Let X be a locally compact Hausdorff topological space
 X is compact $\Leftrightarrow \forall Z, \pi_Z: X \times Z \rightarrow Z$ is closed
(i.e. $\pi_Z(W) \subseteq Z$ is closed, $\forall$ closed $W \subseteq X \times Z$)

Def: A variety X is complete if $\forall$ variety Z the projection map $\pi_Z: X \times Z \rightarrow Z$ is a closed mapping in the Zariski topology

Remark: Any projective variety is complete

Def: (proper map) - above

prop: $f: X \rightarrow Y$ be a continuous mapping of locally compact first countable Hausdorff spaces. Then TFAE:

- (a) f is proper
- (b) f is a closed mapping, i.e. $f(W) \subseteq Y$ is closed, $\forall$ closed subset $W \subseteq X$ and all fibers $f^{-1}(y)$, $y \in Y$ are compact
- (c) $\forall x_k \in X$, s.t. $f(x_k) \in Y$ converges in Y has a subsequence x_{k_l} that converges in X

prop: $f: X \rightarrow Y$ be a continuous map between locally compact Hausdorff space.

(a) f is proper

$\Leftrightarrow$ (b) f is universally closed, i.e.

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \pi_X \uparrow & \circlearrowleft & \uparrow g \text{ continuous mapping} \\ X \times_Y Z & \xrightarrow{\pi_Z} & Z \end{array}$$

the projection π_Z defined by the commutative diagram is a closed mapping

Def: ~~prop~~: A morphism of varieties $\phi: X \rightarrow Y$ is proper if it is universally closed i.e.

$\forall$ variety Z and morphism $\psi: Z \rightarrow Y$ the projection π_Z is a closed mapping in the zariski topology

$$\begin{array}{ccc} X \times_Y Z & \xrightarrow{\pi_X} & X \\ \pi_Z \downarrow & & \downarrow \phi \\ Z & \xrightarrow{\psi} & Y \end{array}$$

Remark: a variety X is complete $\Leftrightarrow \phi: X \rightarrow \{\text{pt}\}$ is proper

Pf: " $\Rightarrow$ " X complete $\Rightarrow \phi: X \times_Y Z \xrightarrow{\pi_Z} Z$ is a closed mapping

" $\Leftarrow$ " $\checkmark$

Furthermore, if X is complete

$\Rightarrow \pi_Z: X \times Z \rightarrow Z$ is proper, $\forall Z$

proof of prop:

" $\Rightarrow$ " $|\Delta'| \subseteq \phi^{-1}(|\Delta|)$ $\checkmark$

We only need to prove $\phi^{-1}(|\Delta|) \subseteq |\Delta'|$

Suppose there exist $v \in |\Delta|$, $v = \phi(v')$ but $v' \notin |\Delta'|$.

Then $\lim_{z \rightarrow 0} \phi_+(\lambda_{v'}(z)) = \lambda_v(z)$ ~~has a limit~~ exists in $X(\Delta)$

but $\lim_{z \rightarrow 0} \lambda_{v'}(z)$ doesn't exist,

which contradicts to properness.

" $\Leftarrow$ " (valuative criterion of properness)

A morphism $f: X \rightarrow Y$ of varieties
(or a separated morphism of
schemes of finite type)

is proper $\Leftrightarrow \forall$ DVR R , with quotient field K , the commutative diagram can be filled by

$$\begin{array}{ccc} \text{Spec}(K) & \xrightarrow{\quad} & X \\ \downarrow & \nearrow & \downarrow \\ \text{Spec}(R) & \xrightarrow{\quad} & Y \end{array}$$

When X is irreducible, one may assume the image of $\text{Spec}(k) \rightarrow X$ is in a given open subset U of X .

Apply this criterion

$$\begin{array}{ccc} \text{Spec}(k) & \rightarrow & X(\Delta') \\ \downarrow & & \downarrow \varphi_+ \\ \text{Spec}(R) & \rightarrow & X(\Delta) \end{array}$$

Assume the image of $\text{Spec}(k) \rightarrow X(\Delta')$ is in T_N'

We have the diagram and consider each affine piece U_0 of $X(\Delta)$

$$\begin{array}{ccc} k & \leftarrow & [M'] \\ \uparrow & & \uparrow \\ R & \leftarrow & [S_6] \end{array}$$

To fill this diagram, we only need to find $6'$ and a map $6' \rightarrow 6$ to fill the following diagram

$$\begin{array}{ccccc} k & \leftarrow & [M'] & \rightarrow & [S_{6'}] \\ \uparrow & \swarrow & \uparrow & \searrow & \uparrow \\ R & \leftarrow & [S_6] & \rightarrow & \end{array} \quad \text{(generating set of } k)$$

Let $\alpha: M' \rightarrow k^*$ is given by $\text{Spec}(k) \rightarrow T_N'$

Then $\forall v \in 6^v \cap M$, $\text{ord} \circ \alpha \circ \varphi^*(m) \geq 0$

$$\Rightarrow \varphi(\text{ord} \circ \alpha) = \text{ord} \circ \alpha \circ \varphi^* \in (6^v)^\vee = 6$$

Since $|\Delta'| = \varphi^{-1}(|\Delta|)$, $\exists 6' \in \Delta'$, s.t.

$$\text{ord} \circ \alpha \in 6' \text{ and } \varphi(6') \subset 6$$

Then $\text{ord} \circ \alpha \in (6^v)^\vee$, so we fill the diagram

Jacobson radical of a ring

- In commutative case

$$J(R) = \bigcap_{M \in \text{Specm} R} M$$

- In Noncommutative case, R a ring with unity
 $J(R) = \{ r \in R \mid rM = 0 \text{ where } M \text{ is simple } R\text{-module} \}$

Motivation

- Example: local ring $\xRightarrow{(R, P)} J(R) = P$
 Artin ring $\xRightarrow{R}$

local ring

- a commutative local ring often arises as the result of the localization of a ring at a prime ideal
- R is a local ring if it has any one of following properties
 - R has a unique maximal left (or right) ideal
 - $1 \neq 0$ and the sum of two non-units in R is a non-unit
 (i.e. the set of non-units in R forms a (proper) ideal)
 - $1 \neq 0$ and if x or $1-x$ is a unit $(\forall x \in R)$
 (i.e. there do not exist two coprime proper ideals $\nexists I_1, I_2 \subsetneq R$ s.t. $R = I_1 + I_2$)
- Example: discrete valuation ring and not a field (which is a local principal ideal domain)

Valuation ring & discrete valuation ring

• Def: Valuation ring D

the following are equivalent

(a) $\forall$ non-zero $x \in K$, at least one
 x or $x^{-1} \in D$ (K : the field of

(b) The ideals of D (fraction)
are totally ordered
by inclusion

(c) There is a totally ordered abelian group
 T and a valuation $v: K \rightarrow T \cup \{\infty\}$
with $D = \{x \in K, v(x) \geq 0\}$

• Def: Discrete ... if $T \cong \mathbb{Z}$

Example of DVR: $F[[X]]$ (F field)